

MediLens AI

AI Diagnostic & Insight System

Model Performance Report

Project Title

MediLens AI - AI Diagnostic & Insight System

Artificial Intelligence Major Project

Submitted By:

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Internship Program:

SkillFlow Artificial Intelligence Internship

Role:

AI Engineer

Domain:

Healthcare Artificial Intelligence

Project Links

GitHub Repository:

https://github.com/Sam-Meera/MediLens_AI_Project

Live Streamlit Demo:

<https://medilensaiprojectgita720329faf7ad5main-main-4fkevgufcra6vpdm8.streamlit.app/>

Technologies Used

- Python
 - NumPy
 - Pandas
 - Scikit-learn
 - TensorFlow/Keras
 - OpenCV
 - Streamlit
 - SHAP
 - HuggingFace Transformers
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1. Project Information

Project Description

MediLens AI is an Artificial Intelligence-based healthcare assistant designed to analyze patient health information and medical images to provide predictive healthcare insights.

The system combines Machine Learning, Deep Learning, Explainable AI, and Natural Language Processing techniques to create an end-to-end AI diagnostic prototype.

The project focuses on assisting healthcare professionals by providing AI-driven risk prediction and medical image analysis.

Project Objective

The main objectives of MediLens AI are:

- To predict chronic disease risk using patient health data.
 - To classify medical images using deep learning models.
 - To compare different AI algorithms and evaluate performance.
 - To provide explainable predictions using XAI techniques.
 - To generate simple understandable summaries of AI results.
 - To deploy trained models through an interactive Streamlit application.
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Role

Role: AI Engineer

Responsibilities performed:

- Data preprocessing and cleaning
 - Exploratory Data Analysis
 - Statistical analysis
 - Machine Learning model development
 - Deep Learning model training
 - Transfer Learning implementation
 - Model evaluation
 - Explainable AI implementation
 - Streamlit deployment
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2. Project Overview

Healthcare data contains valuable information that can help identify disease risks at an early stage.

MediLens AI was developed as a prototype healthcare intelligence system that uses Artificial Intelligence to analyze patient information and medical images.

The project follows the complete AI lifecycle:

1. Data Collection
 2. Data Preprocessing
 3. Exploratory Data Analysis
 4. Model Development
 5. Model Evaluation
 6. Explainability Analysis
 7. Deployment
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3. Technologies and Tools Used

Programming Language

Python

Used for:

- Data processing
 - Model training
 - Application development
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Data Processing Libraries

NumPy

Used for:

- Numerical computations
- Array operations

Pandas

Used for:

- Dataset loading
 - Data cleaning
 - Data analysis
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Machine Learning Frameworks

Scikit-learn

Used for:

- Logistic Regression
 - Random Forest
 - Model evaluation
 - Performance metrics
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Deep Learning Frameworks

TensorFlow/Keras

Used for:

- CNN model development
 - Transfer Learning
 - Medical image classification
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Visualization Tools

- Matplotlib
- Seaborn

Used for:

- Data visualization
 - Correlation analysis
 - Performance comparison
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Deployment Framework

Streamlit

Used for:

- Creating interactive healthcare application
 - Real-time prediction interface
 - Model deployment
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4. Project Workflow

Step 1: Data Collection

Datasets used:

Diabetes Dataset

Used for:

- Patient health risk prediction

Features include:

- Pregnancies
 - Glucose level
 - Blood Pressure
 - Skin Thickness
 - Insulin
 - BMI
 - Diabetes Pedigree Function
 - Age
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Chest X-ray Dataset

Used for:

- Medical image classification
 - Deep learning experiments
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Step 2: Data Preprocessing

Performed preprocessing operations:

- Data cleaning
 - Handling missing values
 - Feature selection
 - Data scaling
 - Image resizing
 - Image normalization
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Step 3: Exploratory Data Analysis

EDA was performed to understand:

- Data distribution
- Feature relationships
- Correlation between variables
- Outliers

Visualization techniques used:

- Histograms
 - Correlation heatmaps
 - Feature distribution plots
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5. AI Models Implemented

Machine Learning Models

Logistic Regression

Purpose:

Used as a baseline classification model for diabetes risk prediction.

Advantages:

- Simple and interpretable
 - Provides probability-based predictions
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Random Forest

Purpose:

Used as an ensemble learning model for comparison.

Advantages:

- Handles complex relationships
- Reduces overfitting

Deep Learning Models

Convolutional Neural Network (CNN)

Purpose:

Used for medical image classification.

CNN automatically learns important image features and classifies medical scans.

MobileNet Transfer Learning

Purpose:

A pre-trained deep learning model was used to improve performance with limited medical image data.

Advantages:

- Faster training
 - Better feature extraction
 - Reduced computational requirements
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6. Model Performance Results

Machine Learning Performance

Model	Accuracy	Precision	Recall	F1 Score
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Logistic Regression	75.32%	64.91%	67.27%	66.07%
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Random Forest	72.73%	-	-	61.82%
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Deep Learning Performance

Model	Accuracy
CNN	89.26%
MobileNet Transfer Learning	87.98%

7. Evaluation Metrics

Accuracy

Accuracy represents the percentage of correctly predicted samples.

Precision

Precision measures the correctness of positive predictions.

Recall

Recall measures how many actual positive cases were identified correctly.

F1 Score

F1 Score provides a balance between Precision and Recall.

8. Explainable AI (XAI)

SHAP Implementation

SHAP (SHapley Additive exPlanations) was used to explain machine learning predictions.

The purpose of SHAP is to understand:

- Which features influence predictions
- How much each feature contributes
- Why a model generated a specific output

Benefits:

- Improves transparency
 - Reduces black-box behaviour
 - Supports responsible AI development
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9. NLP and Language Understanding

Natural Language Processing techniques were explored for medical text understanding.

Implemented concepts:

- Text preprocessing
- Tokenization
- Feature extraction
- Medical text analysis

The goal is to generate understandable explanations from technical AI outputs.

10. Deployment

The trained AI models were deployed using Streamlit.

The application provides:

Features

- Patient information input
 - Diabetes risk prediction
 - Probability score
 - Risk classification
 - Plain English explanation
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Deployment Workflow

User Input



Data Processing



Trained AI Model



Prediction Generation



Result Explanation

11. Ethics and Bias Considerations

Healthcare AI systems must be developed responsibly because predictions can influence medical decisions.

Potential Challenges

- Dataset imbalance
 - Limited diversity in training data
 - Possibility of incorrect predictions
 - Over-dependence on AI systems
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Mitigation Strategies

To improve responsible AI usage:

- Evaluated models using multiple metrics
- Compared multiple algorithms
- Used Explainable AI techniques
- Designed the system as a decision-support tool
- Recommended professional medical consultation

AI predictions should not replace qualified healthcare professionals.

12. Future Enhancements

Future improvements include:

- Multi-disease prediction system
 - More medical image classification models
 - Patient history management
 - Secure authentication
 - Cloud-based deployment
 - Advanced medical report summarization
 - Real-time healthcare monitoring
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13. Conclusion

MediLens AI demonstrates an end-to-end Artificial Intelligence healthcare solution by combining Machine Learning, Deep Learning, Explainable AI, and Web Deployment.

The project successfully implements disease risk prediction, medical image classification, model evaluation, and AI-powered healthcare insights.

The system highlights the potential of Artificial Intelligence in supporting early healthcare decision-making while maintaining transparency and ethical responsibility.